Western Blot Analysis - Experiment Jan 26

2026-02-25

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1 First Western Blot - Axon Compartment

1.1 Blot Images

Channel 680 nm: panTau | **Channel 800 nm:** Actin Note: Lane 1 (Ctrl) was excluded from axon analysis due to unreliable reading.

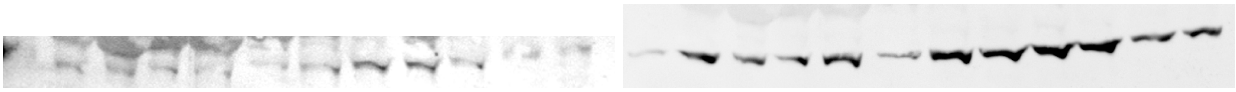
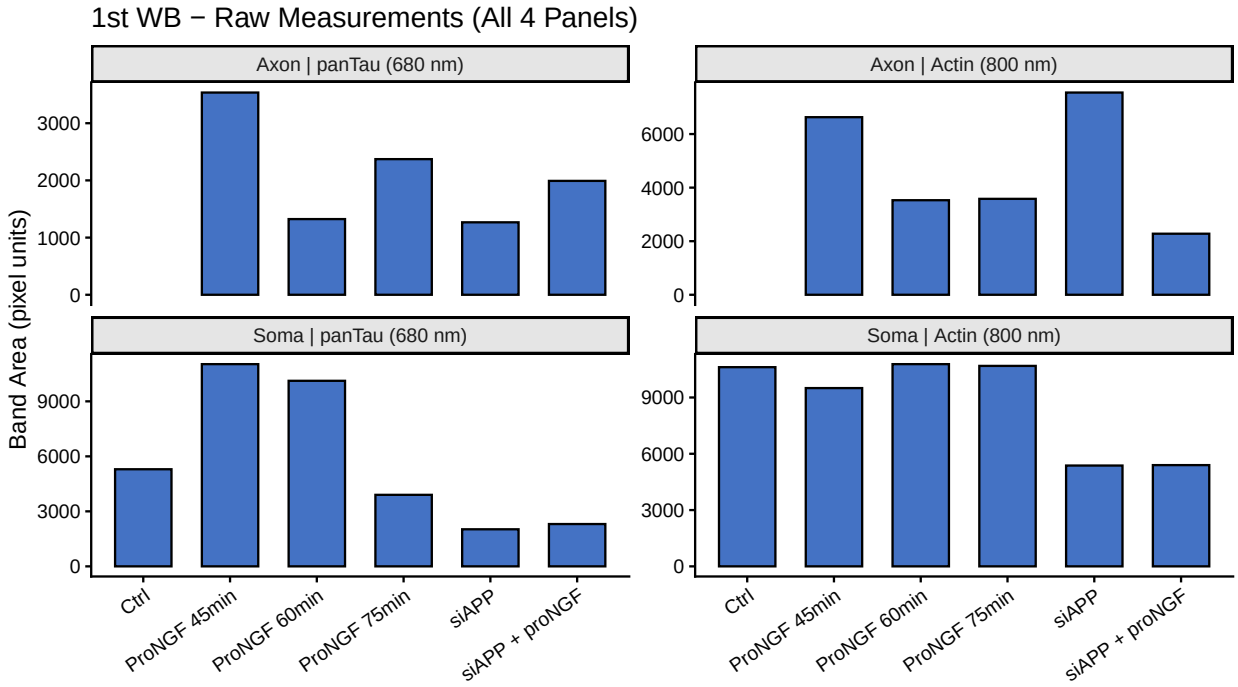
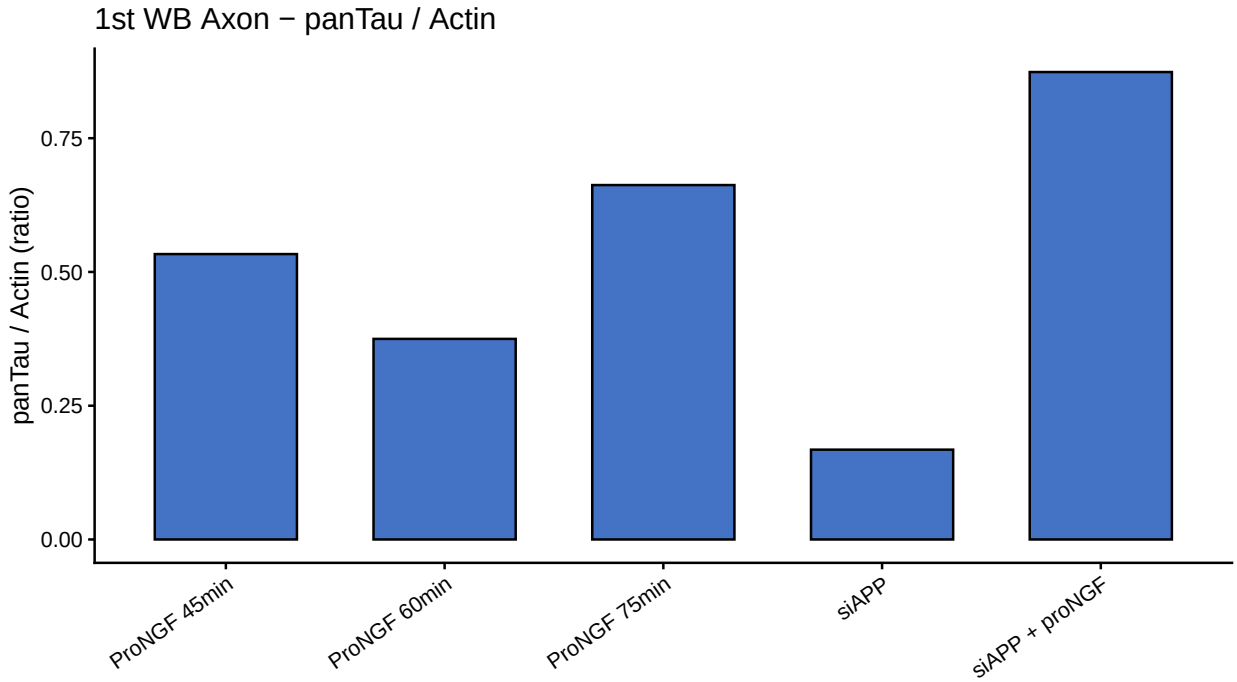


Figure 1: Left: 680 nm panTau | Right: 800 nm Actin

1.2 Raw Measurements - Axon and Soma



1.3 panTau / Actin Ratio - Axon



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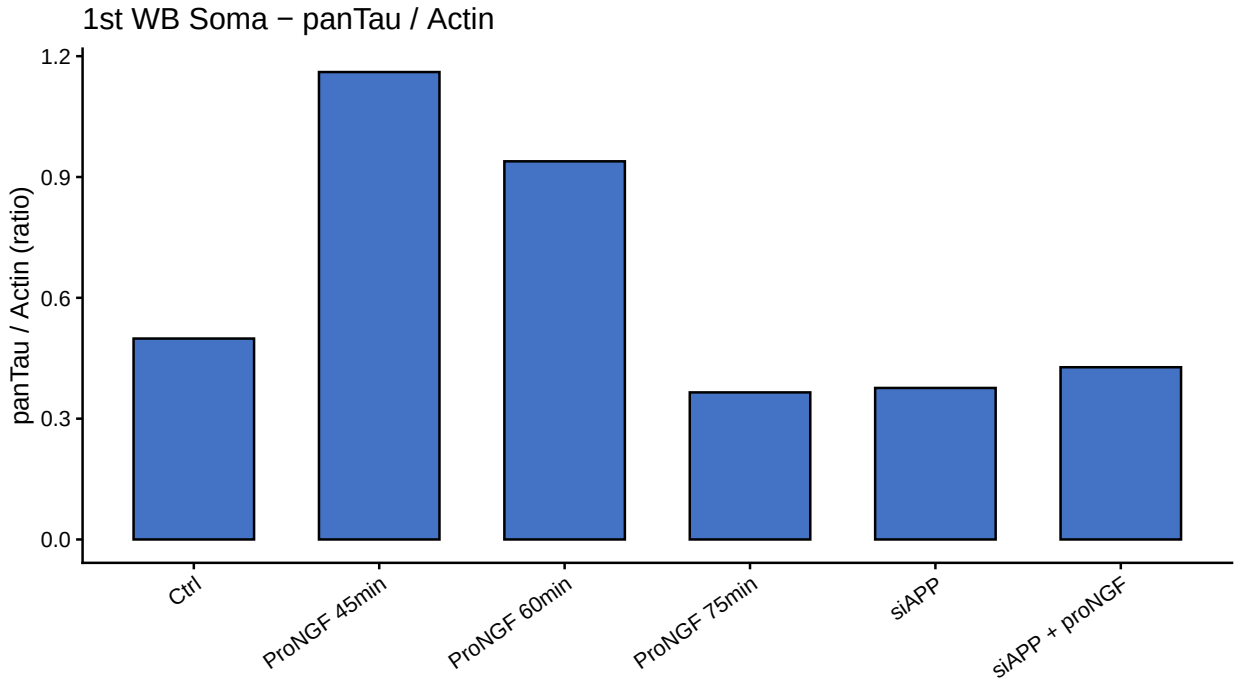
2 First Western Blot - Soma Compartment

Channel 680 nm: panTau | **Channel 800 nm:** Actin

2.1 panTau / Actin Ratio - Soma



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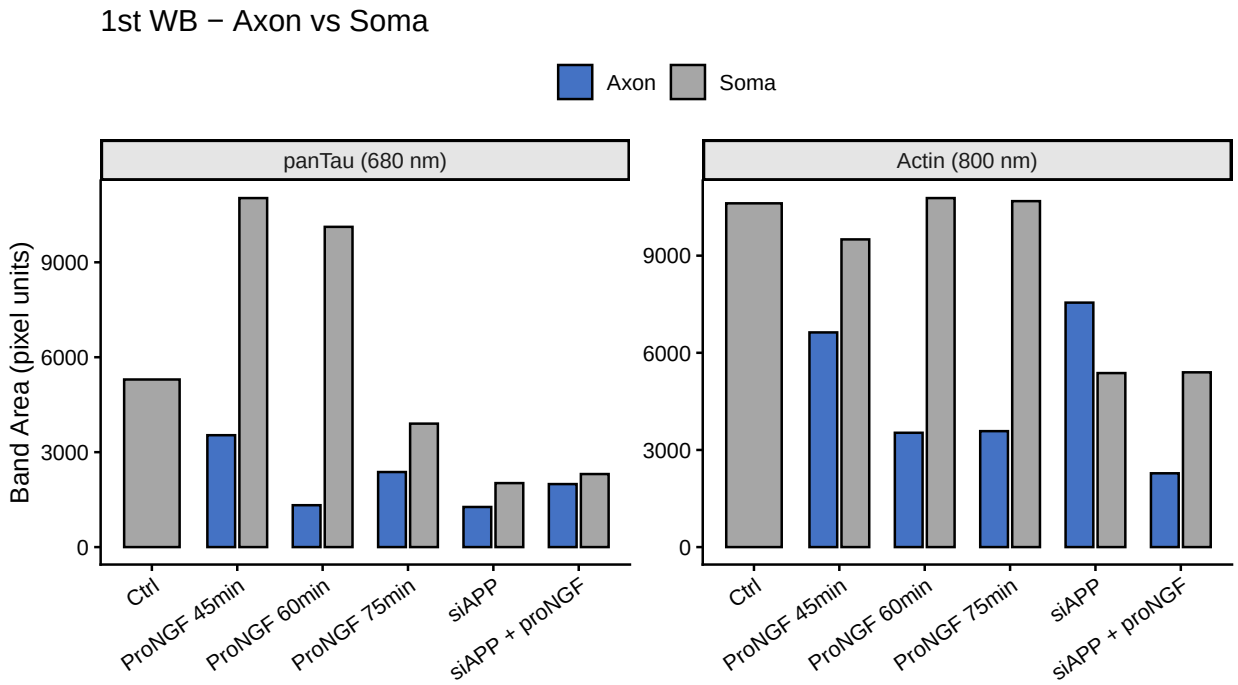
3 First Western Blot - Axon vs Soma Comparison

Axon and Soma measurements side-by-side for both Tau and Actin. Note: Ctrl is soma-only (axon lane 1 excluded).



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4 Second Western Blot - Soma Only

4.1 Blot Images

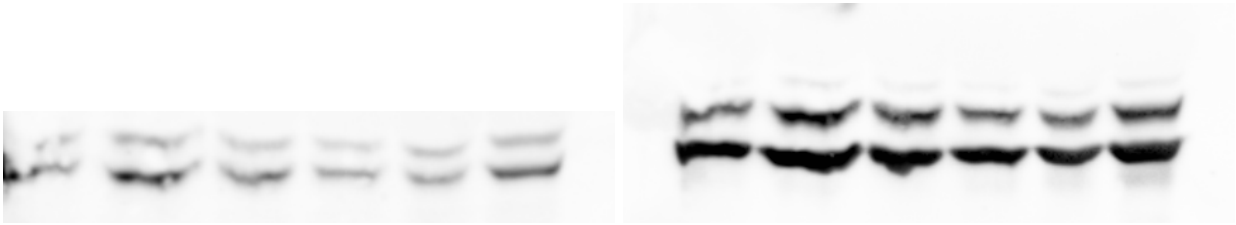
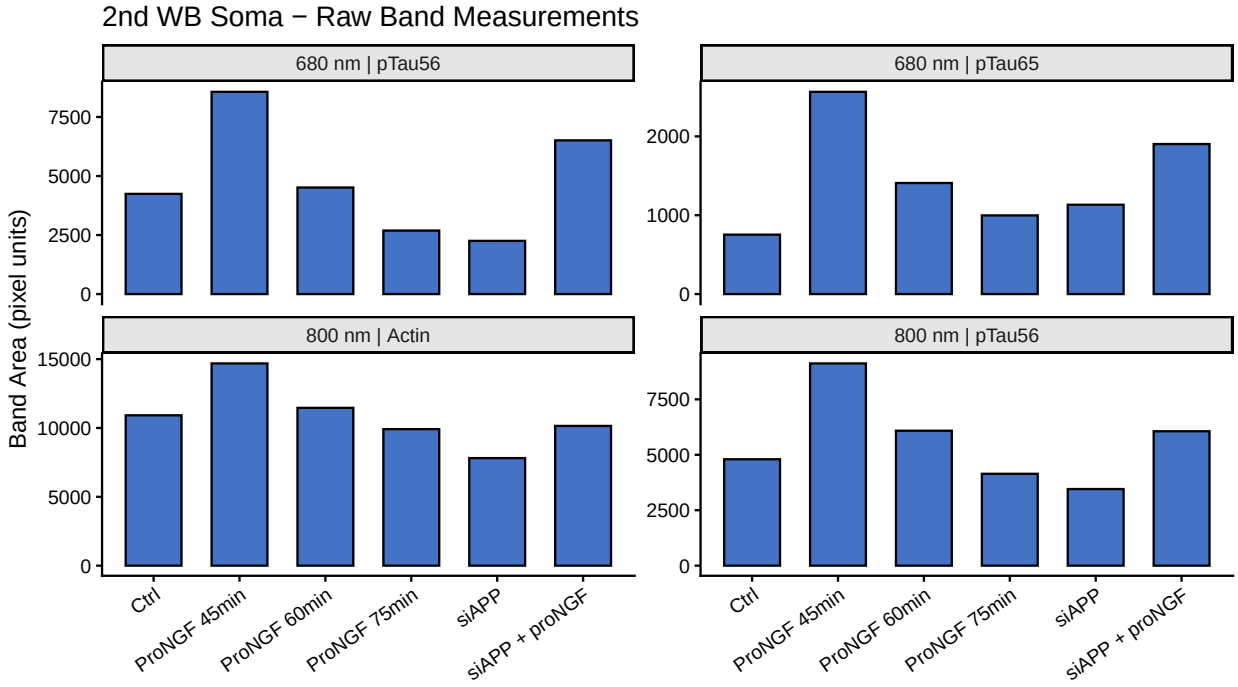


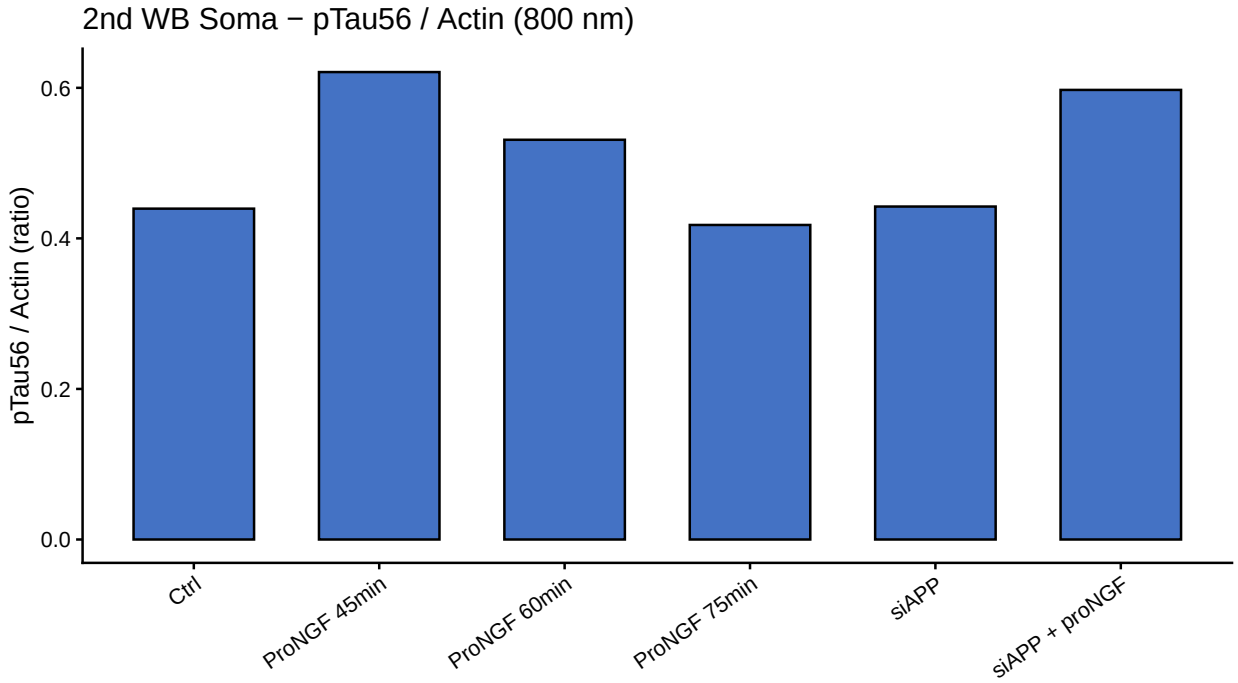
Figure 2: Left: 680 nm pTau65+pTau56 | Right: 800 nm pTau56+Actin

Channel 680 nm: pTau65 (band 1) + pTau56 (band 2) | **Channel 800 nm:** pTau56 (band 1) + Actin (band 2)

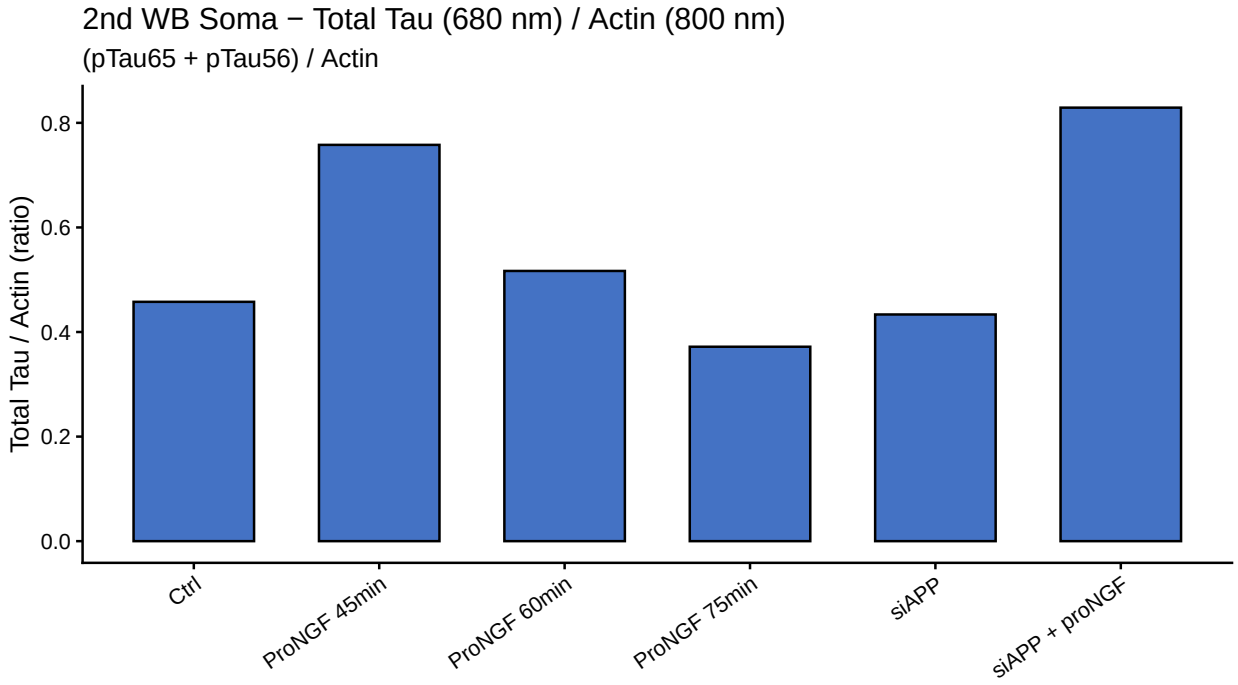
4.2 Raw Band Measurements (2nd WB)



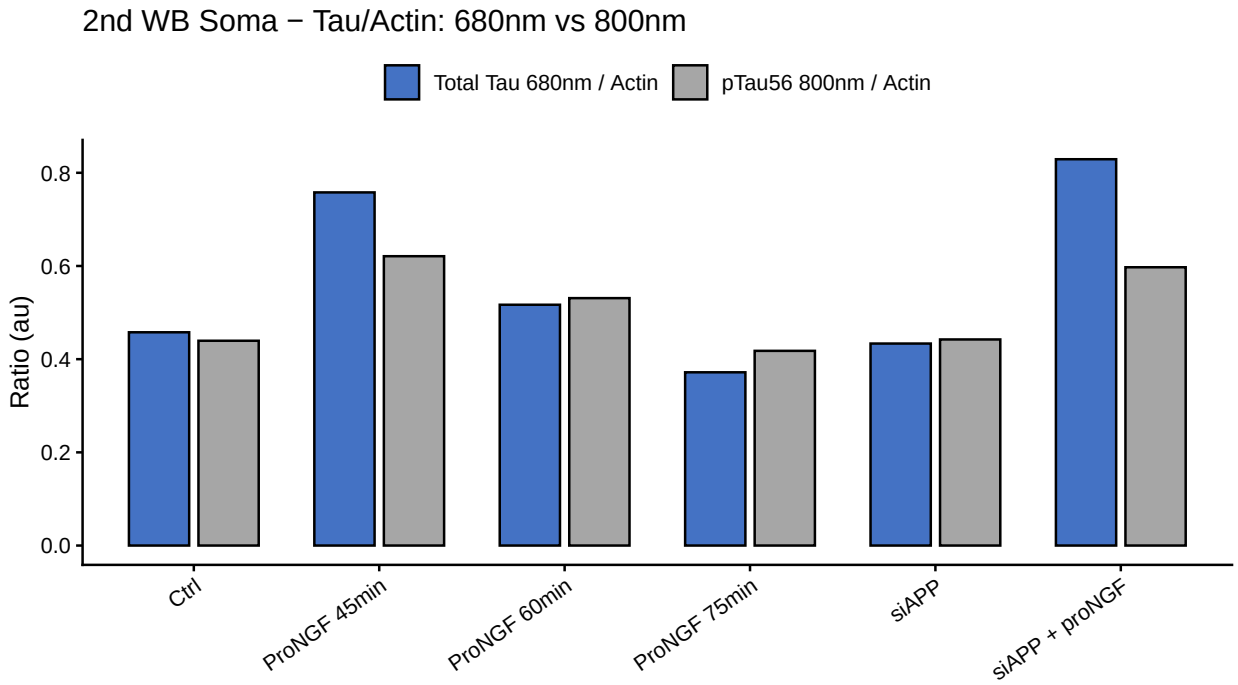
4.3 pTau56 / Actin (800 nm)



4.4 Total Tau (680 nm) / Actin (800 nm)



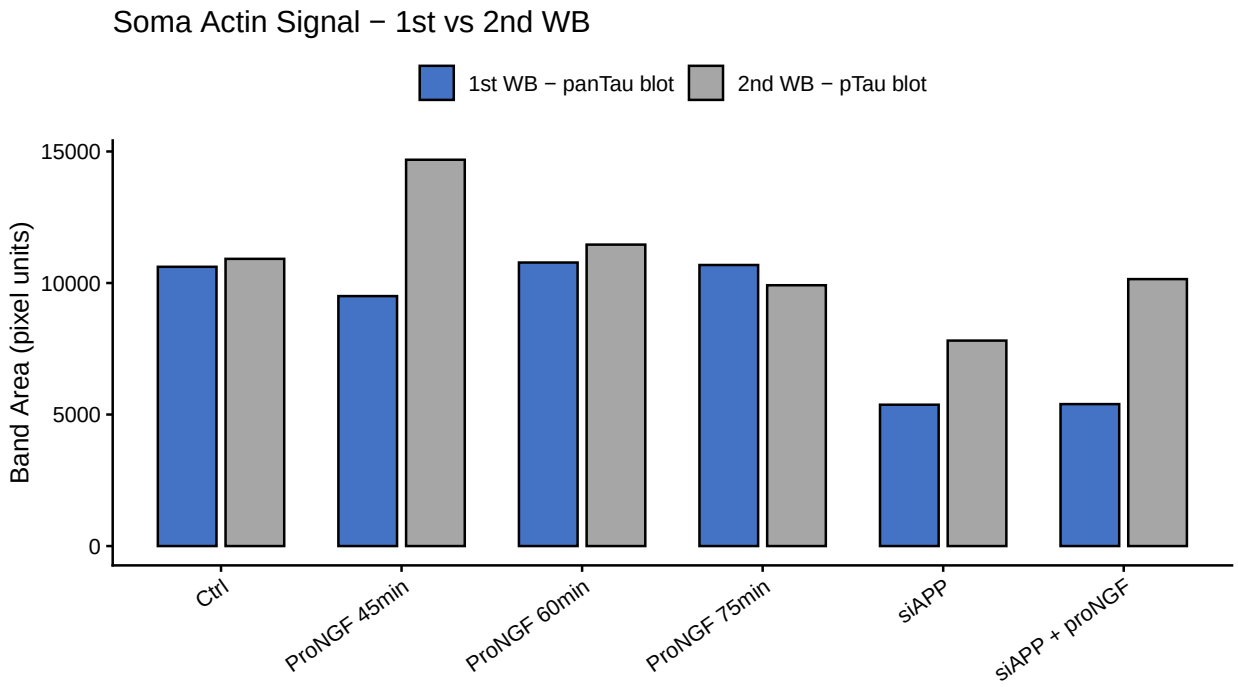
4.5 Both Tau/Actin Ratios Side-by-Side (2nd WB)



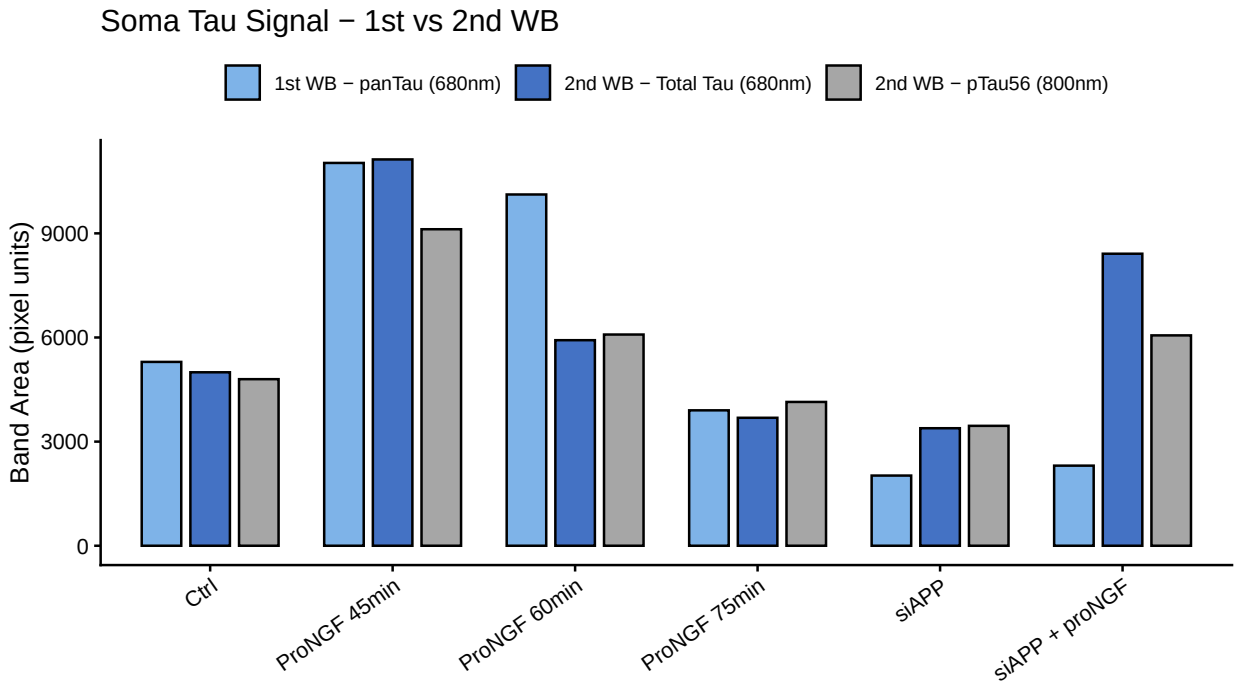
5 Cross-Experiment Comparison - Soma Only

Comparing 1st WB (panTau/Actin) vs 2nd WB (Total Tau/Actin and pTau56/Actin) for all conditions present in both experiments.

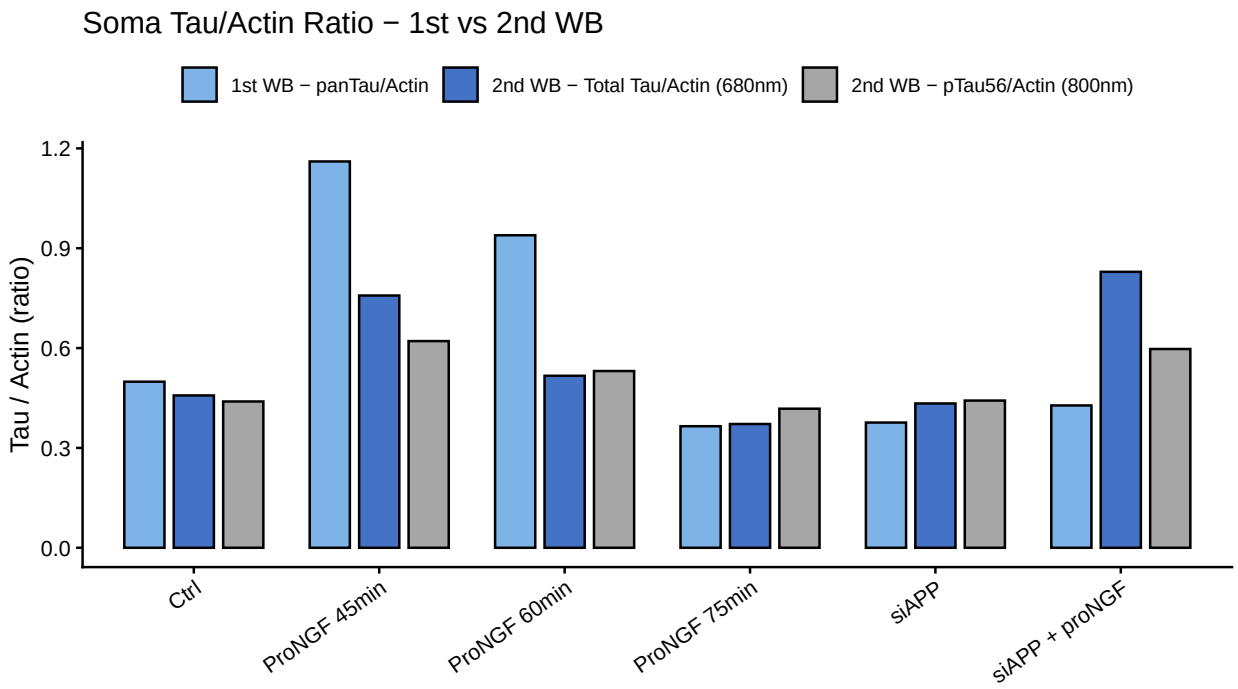
5.1 Actin Signal Comparison



5.2 Tau Signal Comparison



5.3 Tau / Actin Ratio Comparison



6 Session Info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.0
##  [5] purrr_1.2.1     readr_2.1.6     tidyr_1.3.2     tibble_3.3.1
##  [9] ggplot2_4.0.2   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0       yaml_2.3.10
##  [9] fastmap_1.2.0      R6_2.6.1           labeling_0.4.3     generics_0.1.4
## [13] knitr_1.51         bookdown_0.46      pillar_1.11.1      RColorBrewer_1.1-3
## [17] tzdb_0.5.0         rlang_1.1.7        stringi_1.8.7      xfun_0.56
## [21] S7_0.2.1           bit64_4.6.0-1      otel_0.2.0         timechange_0.4.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.4     digest_0.6.37
## [29] grid_4.5.1         vroom_1.7.0        rstudioapi_0.18.0  hms_1.1.4
## [33] lifecycle_1.0.5    vctrs_0.7.1        evaluate_1.0.5     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.30     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.9
```